# TEAS 7 Science Review Sheets

Compact review of the four TEAS 7 Science sub-areas: human anatomy & physiology, biology, chemistry and scientific reasoning

MindShelf

## The Science Section at a Glance

| Sub-area (ATI TEAS 7 content outline) | Scored items |
|---|---|
| **Human anatomy & physiology**: Body orientation; respiratory, cardiovascular, digestive, nervous, muscular, skeletal, reproductive, integumentary, endocrine, urinary and immune systems | 18 |
| **Biology**: Cells; DNA, RNA and proteins; Mendelian inheritance; macromolecules; microorganisms and disease | 9 |
| **Chemistry**: Atomic structure; properties and changes of matter; reactions and reaction conditions; solutions; acids and bases | 8 |
| **Scientific reasoning**: Measurements and tools; logic and evidence; relationships between variables; the scientific method | 9 |

44 scored items plus 6 unscored pretest items (50 in all) in 60 min. Pretest items look like the others, so treat every question as scored. These sheets review core facts for each sub-area; they are not a complete course and contain no actual test questions.

## Human Anatomy & Physiology

Largest sub-area: 18 scored items.

### Orientation

**Anatomical position:** standing upright, facing forward, arms at the sides, palms forward, feet parallel. **Supine** = lying face up; **prone** = face down. Left and right always mean the *patient*'s left and right.

| Term | Opposite | Meaning |
|---|---|---|
| **Superior (cranial)** | Inferior (caudal) | Toward the head / toward the feet |
| **Anterior (ventral)** | Posterior (dorsal) | Front / back of the body |
| **Medial** | Lateral | Toward / away from the midline |
| **Proximal** | Distal | Closer to / farther from the trunk or a limb's attachment |
| **Superficial** | Deep | Toward / away from the body surface |

| Plane | Divides the body into |
|---|---|
| **Sagittal (midsagittal = equal halves)** | Right and left parts |
| **Frontal (coronal)** | Anterior and posterior parts |
| **Transverse (horizontal)** | Superior and inferior parts |

**Cavities:** dorsal = cranial (brain) + vertebral (spinal cord); ventral = thoracic (pleural cavities, pericardial cavity, mediastinum) + abdominopelvic, separated by the diaphragm.

**Levels of organization:** chemical → cell → tissue → organ → organ system → organism. Four tissue types: epithelial, connective, muscle, nervous.

**Homeostasis:** negative feedback reverses a change (body temperature near 37 °C / 98.6 °F, blood glucose, blood pH 7.35–7.45); positive feedback amplifies it until an end point (oxytocin in labor, blood clotting).

### Cardiovascular System

| | |
|---|---|
| **Right atrium** | Receives deoxygenated blood from the superior and inferior venae cavae |
| **Right ventricle** | Pumps blood through the pulmonary valve into the pulmonary trunk (pulmonary circuit) |
| **Left atrium** | Receives oxygenated blood from the pulmonary veins |
| **Left ventricle** | Thickest wall; pumps blood through the aortic valve into the aorta (systemic circuit) |
| **Valves** | Tricuspid (right) and mitral or bicuspid (left); semilunar valves (pulmonary and aortic) guard the outflow; all prevent backflow |
| **Conduction** | SA node (pacemaker) → AV node → AV bundle (bundle of His) → bundle branches → Purkinje fibers |
| **Vessels** | Arteries carry blood away from the heart; capillaries (one cell thick) exchange gases and nutrients; veins return blood, with valves in the limbs |
| **Blood** | Erythrocytes carry $O_2$ on hemoglobin; leukocytes defend; platelets start clotting; plasma carries solutes |

4 chambers. Blood path: venae cavae → right atrium → tricuspid → right ventricle → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral → left ventricle → aorta → body.

**Cardiac output**

$$CO = HR \times SV$$

*E.g.* HR 70/min, SV 70 mL: CO = 70 × 70 mL/min = 4.9 L/min.

### Respiratory System

| | |
|---|---|
| **Nasal cavity** | Warms, humidifies and filters inhaled air |
| **Pharynx and larynx** | Shared airway and food passage; larynx holds the vocal cords; the epiglottis covers it during swallowing |
| **Trachea and bronchi** | Cartilage rings keep the trachea open; bronchi branch into bronchioles (conducting zone) |
| **Alveoli** | Gas exchange by diffusion across a thin respiratory membrane; surfactant lowers surface tension |
| **Diaphragm** | Contracts and flattens on inspiration: chest volume up, pressure down, air flows in |
| **Brainstem centers** | Medulla and pons set the breathing rhythm; rising $CO_2$ (falling pH) is the main drive to breathe |

Right lung 3 lobes, left lung 2 lobes (room for the heart). $O_2$ and $CO_2$ move by diffusion, from higher to lower partial pressure.

### Digestive System

| | |
|---|---|
| **Mouth** | Chewing (mechanical digestion); salivary amylase starts starch digestion |
| **Esophagus** | Moves the bolus to the stomach by peristalsis |
| **Stomach** | HCl and pepsin begin protein digestion; churning makes chyme |
| **Small intestine** | Duodenum, jejunum, ileum: most chemical digestion and absorption; villi and microvilli enlarge the surface |
| **Liver and gallbladder** | Liver makes bile (emulsifies fats); gallbladder stores and concentrates it |
| **Pancreas (exocrine)** | Releases amylase, lipase, proteases and bicarbonate into the duodenum |
| **Large intestine** | Absorbs water and electrolytes; gut bacteria make vitamin K; forms feces |

Order: mouth → pharynx → esophagus → stomach → small intestine → large intestine → rectum → anus. Liver, gallbladder and pancreas are accessory organs: food never passes through them.

### Nervous System

| | |
|---|---|
| **CNS** | Brain and spinal cord: integrate information |
| **Neuron** | Dendrites receive, the cell body integrates, the axon conducts; myelin (Schwann cells in the PNS, oligodendrocytes in the CNS) speeds conduction |
| **Cerebrum** | Lobes: frontal (motor, planning, personality), parietal (touch, body sense), temporal (hearing, memory), occipital (vision) |
| **Cerebellum** | Coordination, balance, posture |
| **Brainstem** | Midbrain, pons, medulla oblongata: breathing, heart rate, blood pressure |
| **Autonomic NS** | Sympathetic: fight or flight (heart rate up, pupils dilate, airways dilate, digestion slows); parasympathetic: rest and digest |
| **Action potential** | $Na^+$ rushes in (depolarization), $K^+$ flows out (repolarization); neurotransmitters such as acetylcholine cross the synapse |

PNS: 12 pairs of cranial nerves, 31 pairs of spinal nerves; afferent (sensory) toward the CNS, efferent (motor) away from it.

### Muscular & Skeletal Systems

| | |
|---|---|
| **Skeletal muscle** | Voluntary, striated, multinucleated fibers |
| **Cardiac muscle** | Involuntary, striated, branched cells joined by intercalated discs |
| **Smooth muscle** | Involuntary, not striated; walls of hollow organs and blood vessels |
| **Sarcomere** | Contraction unit: actin (thin) and myosin (thick) filaments slide past each other; needs $Ca^{2+}$ and ATP |
| **Bone functions** | Support, protection, movement, mineral ($Ca^{2+}$) storage, blood cell formation in red marrow |
| **Bone cells** | Osteoblasts build bone, osteoclasts break it down, osteocytes maintain it |
| **Connections** | Tendons join muscle to bone; ligaments join bone to bone; synovial joints move freely |

Adult skeleton: 206 bones. Axial = skull, vertebral column, rib cage; appendicular = limbs and their girdles. Vertebrae: 7 cervical, 12 thoracic, 5 lumbar, then sacrum and coccyx.

### Endocrine System

| | |
|---|---|
| **Hypothalamus** | Releasing and inhibiting hormones control the anterior pituitary; makes ADH and oxytocin |
| **Anterior pituitary** | GH, TSH, ACTH, FSH, LH, prolactin |
| **Posterior pituitary** | Releases ADH (water reabsorption) and oxytocin (uterine contraction, milk ejection) |
| **Thyroid** | $T_4$ and $T_3$ raise metabolic rate; calcitonin lowers blood $Ca^{2+}$ |
| **Parathyroids** | PTH raises blood $Ca^{2+}$ |
| **Adrenal cortex** | Cortisol (stress, raises glucose); aldosterone ($Na^+$ and water retention) |
| **Adrenal medulla** | Epinephrine and norepinephrine (fight or flight) |
| **Pancreatic islets** | Insulin from beta cells lowers blood glucose; glucagon from alpha cells raises it |
| **Pineal gland** | Melatonin (sleep–wake rhythm) |

Most hormone levels are held steady by negative feedback (e.g., high $T_4$ suppresses TSH).

### Urinary System

| | |
|---|---|
| **Kidneys** | Filter blood; regulate water, electrolytes, pH and blood pressure; release renin and erythropoietin (EPO) |
| **Nephron** | Glomerulus and glomerular (Bowman's) capsule filter → proximal tubule (most reabsorption) → nephron loop (loop of Henle) → distal tubule → collecting duct |

Written with AI assistance; content checked and approved by the seller. · A study aid. It complements your courses and reference books; it doesn't replace them. · TEAS is a registered trademark of ATI (Assessment Technologies Institute). Not affiliated with or endorsed by ATI.

| | |
|---|---|
| **Hormonal control** | ADH increases water reabsorption in collecting ducts; aldosterone increases $Na^+$ reabsorption |
| **Urine path** | Ureters (peristalsis) → urinary bladder (storage) → urethra |

### Reproductive System

| | |
|---|---|
| **Testes** | Sperm production in seminiferous tubules; testosterone from interstitial cells |
| **Male ducts and glands** | Epididymis (maturation, storage) → ductus (vas) deferens → urethra; seminal vesicles, prostate and bulbourethral glands add semen fluid |
| **Ovaries** | Oocytes; estrogen and progesterone |
| **Uterine tubes and uterus** | Fertilization usually occurs in the uterine (fallopian) tube; implantation in the endometrium, which sheds at menstruation |
| **Hormones** | FSH and LH from the anterior pituitary drive gamete production; the LH surge triggers ovulation |

### Integumentary System

| | |
|---|---|
| **Epidermis** | Keratinized stratified squamous epithelium, no blood vessels; melanocytes make melanin (UV protection) |
| **Dermis** | Connective tissue with blood vessels, nerves, hair follicles, sweat and sebaceous glands |
| **Hypodermis** | Subcutaneous fat: insulation, energy storage, anchoring |
| **Functions** | Barrier, thermoregulation (sweat, blood flow), sensation, vitamin D synthesis |

### Immune & Lymphatic Systems

| | |
|---|---|
| **Innate defenses** | Nonspecific: skin and mucous membranes, phagocytes (neutrophils, macrophages), NK cells, inflammation, fever |
| **B cells** | Humoral immunity: become plasma cells that secrete antibodies; mature in red bone marrow |
| **T cells** | Cell-mediated immunity: helper T cells coordinate, cytotoxic T cells kill infected cells; mature in the thymus |
| **Lymphatic organs** | Lymph nodes filter lymph; spleen filters blood and removes old red cells; lymphatic vessels return fluid to the blood |
| **Immunity types** | Active (infection or vaccine, makes memory cells) vs passive (antibodies received, e.g., across the placenta; short-lived) |

# Biology

9 scored items.

### Cells & Organelles

| Structure | Function |
|---|---|
| **Nucleus** | Holds the DNA; the nucleolus assembles ribosome subunits |
| **Ribosome** | Builds proteins (free in cytosol or bound to rough ER) |
| **Rough ER** | Ribosome-studded; makes proteins for membranes and export |
| **Smooth ER** | Makes lipids, detoxifies, stores $Ca^{2+}$ |
| **Golgi apparatus** | Modifies, sorts and packages proteins into vesicles |
| **Mitochondrion** | Aerobic cellular respiration → ATP; has its own DNA |
| **Lysosome** | Digestive enzymes break down waste and worn-out parts (animal cells) |
| **Cell membrane** | Phospholipid bilayer with proteins (fluid mosaic), selectively permeable |
| **Cytoskeleton** | Microtubules and microfilaments: shape, transport, movement |
| **Centrioles** | Organize the spindle in animal cells |
| **Plant cells only** | Cell wall (cellulose), chloroplasts (photosynthesis), large central vacuole |

**Prokaryotes** (bacteria, archaea): no nucleus, no membrane-bound organelles; DNA in a nucleoid; ribosomes present. **Eukaryotes** (animals, plants, fungi, protists): nucleus and organelles.

**Membrane transport.** Passive, no ATP, down the gradient: diffusion, facilitated diffusion, osmosis (water moves toward the higher solute concentration). Active, uses ATP, against the gradient: e.g., the $Na^+/K^+$ pump, endocytosis, exocytosis.

| Cell placed in | Result |
|---|---|
| **Hypertonic** solution | Water leaves; the cell shrinks (crenates) |
| **Hypotonic** solution | Water enters; the cell swells, may burst (lyse) |
| **Isotonic** solution | No net water movement |

### Cell Division

**Cell cycle:** interphase ($G_1$ growth, S DNA replication, $G_2$ preparation) → mitosis: prophase, metaphase (chromosomes line up), anaphase (sister chromatids separate), telophase (PMAT) → cytokinesis.

| | Mitosis | Meiosis |
|---|---|---|
| **Purpose** | Growth, repair, asexual reproduction | Gametes (eggs and sperm) |
| **Divisions** | One | Two (meiosis I and II) |
| **Daughter cells** | Two, genetically identical | Four, genetically different |
| **Chromosome number** | Diploid (2n) kept | Haploid (n): halved |
| **Variation** | None (barring mutation) | Crossing over (prophase I) and independent assortment |

Humans: 2n = 46 (23 pairs); a gamete has $n = 46/2 = 23$.

### DNA, RNA & Protein Synthesis

| | DNA | RNA |
|---|---|---|
| **Sugar** | Deoxyribose | Ribose |
| **Bases** | A, T, G, C | A, U, G, C |
| **Strands** | Double helix | Usually single |
| **Role** | Stores genetic information | Carries it (mRNA), brings amino acids (tRNA), forms ribosomes (rRNA) |

Base pairing: A–T and G–C in DNA; A–U in RNA. DNA replication is semiconservative (each new helix keeps one old strand).

**Central dogma**

$$\text{DNA} \xrightarrow{\text{transcription (nucleus)}} \text{mRNA} \xrightarrow{\text{translation (ribosome)}} \text{protein}$$

A **codon** is 3 bases of mRNA; the tRNA **anticodon** pairs with it. $4^3 = 64$ codons code for 20 amino acids plus 3 stop codons (UAA, UAG, UGA): the code is redundant. AUG = start (methionine). *E.g.* template DNA TAC CGT → mRNA AUG GCA → Met–Ala.

**Mutations:** substitution (point; may change one amino acid or none); insertion or deletion (shifts the reading frame: frameshift).

### Mendelian Genetics

**Allele:** version of a gene. **Genotype** (AA, Aa, aa) vs **phenotype** (observed trait). Homozygous = two identical alleles; heterozygous = two different. **Law of segregation:** allele pairs separate into gametes. **Independent assortment:** genes on different chromosomes sort independently.

**EXAMPLE Punnett square, Aa × Aa**

| | A | a |
|---|---|---|
| **A** | AA | Aa |
| **a** | Aa | aa |

Genotypes AA : Aa : aa = 1 : 2 : 1 phenotypes dominant : recessive = 3 : 1 P(aa) = 0.5 × 0.5 = 25 %.

- Test cross Aa × aa: P(aa) = 0.5 × 1 = 50 %.
- Dihybrid AaBb × AaBb, out of 16: 9 : 3 : 3 : 1 (e.g., both dominant = 0.75 × 0.75 × 16 = 9).

**Non-Mendelian:** incomplete dominance (blend: red × white → pink); codominance (both expressed: blood type AB); multiple alleles (ABO); X-linked recessive traits (e.g., hemophilia) appear more often in males.

### Macromolecules & Enzymes

| Class | Building block | Examples / role |
|---|---|---|
| **Carbohydrates** | Monosaccharides (e.g. glucose) | Glucose, sucrose, starch, glycogen, cellulose. Quick energy, storage, plant structure |
| **Lipids** | Glycerol + fatty acids (not true polymers) | Triglycerides, phospholipids, steroids (cholesterol). Long-term energy, membranes, hormones, insulation |
| **Proteins** | Amino acids joined by peptide bonds | Enzymes, hemoglobin, collagen, antibodies. Catalysis, structure, transport, defense; shape sets function |
| **Nucleic acids** | Nucleotides (sugar + phosphate + base) | DNA, RNA (ATP is a nucleotide). Store and express genetic information |

Dehydration synthesis joins monomers (releases water); hydrolysis splits them (uses water). **Enzymes** are mostly proteins: they lower activation energy, fit specific substrates at the active site and are not used up; heat and extreme pH denature them.

### Microorganisms & Disease

| Group | Structure | Examples / treatment |
|---|---|---|
| **Bacteria** | Prokaryotic cells; peptidoglycan wall; cocci, bacilli, spirilla | Strep throat, tuberculosis; treated with antibiotics |
| **Viruses** | Not cells: DNA or RNA in a protein capsid; replicate only inside host cells | Influenza, HIV; antibiotics do not work; antivirals, vaccines |
| **Fungi** | Eukaryotic; chitin cell wall; yeasts and molds | Candidiasis, athlete's foot; antifungals |
| **Protozoa** | Single-celled eukaryotes | Malaria (Plasmodium), giardiasis |

**Gram stain:** Gram-positive = thick peptidoglycan wall, stains purple; Gram-negative = thin wall plus outer membrane, stains pink.

# Chemistry

8 scored items.

### Atomic Structure & Periodic Table

| Particle | Charge | Location |
|---|---|---|
| **Proton** | positive | nucleus; their number is the atomic number *Z* and defines the element |
| **Neutron** | none | nucleus |

Written with AI assistance; content checked and approved by the seller. · A study aid. It complements your courses and reference books; it doesn't replace them. · TEAS is a registered trademark of ATI (Assessment Technologies Institute). Not affiliated with or endorsed by ATI.

| Particle | Charge | Location |
|---|---|---|
| **Electron** | negative | shells (energy levels) around the nucleus; valence electrons drive bonding |

Mass number $A = Z + N$; **isotopes** = same $Z$, different $N$; **ion** = atom that gained (anion, −) or lost (cation, +) electrons. *E.g.* carbon-14: $N = 14 - 6 = 8$; chlorine-35: $N = 35 - 17 = 18$; $Na^+$: 11 protons − 10 electrons = charge +1.

**Periodic table:** rows (periods) = number of electron shells; columns (groups) = same valence electrons, similar properties. Metals on the left, nonmetals on the right, metalloids along the staircase; alkali metals (far-left column) are very reactive; noble gases (far-right column) are stable and unreactive. Electronegativity increases toward the upper right (fluorine highest); atomic radius increases toward the lower left.

## Chemical Bonds

| Bond | How | Example |
|---|---|---|
| **Ionic** | Electrons transferred (metal → nonmetal); oppositely charged ions attract | NaCl |
| **Nonpolar covalent** | Electrons shared equally | $H_2$, $O_2$ |
| **Polar covalent** | Electrons shared unequally: partial charges (δ+ / δ−) | $H_2O$ |
| **Hydrogen bond** | Weak attraction between H bonded to N/O/F and a nearby N/O/F | Between water molecules, DNA base pairs |
| **Metallic** | Cations in a sea of shared electrons | Cu, Fe |

## Properties & States of Matter

**Solid:** fixed shape and volume. **Liquid:** fixed volume, takes the container's shape. **Gas:** neither; compressible.

| Phase change | From | To | Heat |
|---|---|---|---|
| **Melting** | solid | liquid | absorbed |
| **Freezing** | liquid | solid | released |
| **Vaporization (evaporation / boiling)** | liquid | gas | absorbed |
| **Condensation** | gas | liquid | released |
| **Sublimation** | solid | gas | absorbed |
| **Deposition** | gas | solid | released |

**Physical change:** same substance (melting, dissolving, cutting). **Chemical change:** new substance (signs: gas, color change, precipitate, heat or light). Water melts at 0 °C and boils at 100 °C (1 atm); ice floats because hydrogen bonding makes it less dense than liquid water; water has a high specific heat, cohesion and adhesion.

## Chemical Reactions

| Type | Pattern | Example |
|---|---|---|
| **Synthesis** | A + B → AB | $2\ H_2 + O_2 \rightarrow 2\ H_2O$ |
| **Decomposition** | AB → A + B | $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$ |
| **Single replacement** | A + BC → AC + B | $Zn + 2\ HCl \rightarrow ZnCl_2 + H_2$ |
| **Double replacement** | AB + CD → AD + CB | $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ |
| **Combustion** | fuel + $O_2$ → $CO_2$ + $H_2O$ | $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ |
| **Neutralization** | acid + base → salt + water | $HCl + NaOH \rightarrow NaCl + H_2O$ |

**Balancing: conservation of mass**
Change coefficients only, never subscripts. $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$: O left 2 × 2 = 4, O right 2 + 2 = 4; H right 2 × 2 = 4, as in $CH_4$.

**Faster reactions:** higher temperature, higher concentration, more surface area, a catalyst (lowers activation energy, not consumed). **Exothermic** releases heat; **endothermic** absorbs it. **Le Chatelier:** a system at equilibrium shifts to offset a change in concentration, temperature or pressure.

## Solutions

Solute dissolves in solvent (water: the "universal solvent"); like dissolves like (polar with polar). Unsaturated → saturated → supersaturated. Most solids dissolve better when hot; gases dissolve better when cold and under pressure.

**Concentration**

$$M = \frac{\text{mol solute}}{\text{L solution}} \quad C_1V_1 = C_2V_2 \quad \%\,(m/v) = \frac{\text{g solute}}{100\ \text{mL solution}}$$

- NaCl molar mass 22.99 + 35.45 = 58.44 g/mol; 14.61 g in 0.5 L: $M$ = 14.61/(58.44 × 0.5) = 0.50 M.
- 500 mL of 0.5 M from 2 M stock: $V_1$ = 0.5 × 500/2 = 125 mL, then add solvent to 500 mL.
- 250 mL of 5 % (m/v) glucose holds 5 × 250/100 = 12.5 g.

## Acids & Bases

**Acid:** donates $H^+$, pH < 7, sour, turns blue litmus red. **Base:** accepts $H^+$ (or releases $OH^-$), pH > 7, bitter, slippery, turns red litmus blue. Neutral = 7 (25 °C); scale 0–14.

$$pH = -\log[H^+] \quad pH + pOH = 14 \quad (25\ °C)$$

Each pH unit is a tenfold change: pH 3 has $10^{6-3}$ = 1,000 times the $[H^+]$ of pH 6. At pH 3, pOH = 14 − 3 = 11.

Strong acids (HCl) ionize fully; weak acids (acetic, carbonic) only partly. **Buffers** resist pH change: the bicarbonate buffer keeps blood pH at 7.35–7.45.

# Scientific Reasoning

9 scored items.

## Scientific Method

Observation → question → **hypothesis** (testable, falsifiable prediction) → controlled experiment → data analysis → conclusion → repeat and communicate.

| Term | Meaning |
|---|---|
| **Independent variable** | What the investigator changes (x-axis) |
| **Dependent variable** | What is measured in response (y-axis) |
| **Controlled variables** | Kept the same in every trial |
| **Control group** | Gets no treatment: the baseline for comparison |
| **Theory** | Broad, well-tested explanation |
| **Law** | Describes what happens (often an equation), not why |

**Accuracy** = closeness to the true value; **precision** = closeness of repeated results to each other. More trials and larger samples make results more reliable. Correlation does not prove causation. An **observation** is what you sense or measure; an **inference** is a conclusion drawn from it. Inductive reasoning goes from specific cases to a general rule; deductive reasoning applies a general rule to a specific case.

**Working with data**
- Mean of trials 12, 15, 18: (12 + 15 + 18)/3 = 15.
- Slope: 4 cm on day 2, 10 cm on day 8: (10 − 4)/(8 − 2) = 1 cm/day.
- Percent error: measured 10.4, accepted 10: (10.4 − 10)/10 × 100 = 4 %.

**Graphs:** line = change over time or a continuous variable; bar = compare categories; pie = parts of a whole; scatter = relationship between two variables (positive, negative or none). Interpolation reads between data points; extrapolation goes beyond them (less reliable).

## Measurement & Units

**Tools:** balance (mass), graduated cylinder (volume: read the bottom of the meniscus at eye level), ruler or meter stick (length), thermometer (temperature), pipette (small volumes). **Mass** (kg, same everywhere) is not **weight** (a force that depends on gravity).

| Prefix | Symbol | × base unit |
|---|---|---|
| **kilo** | k | 1,000 |
| **(base unit)** | — | — |
| **deci** | d | 0.1 |
| **centi** | c | 0.01 |
| **milli** | m | 0.001 |
| **micro** | μ | 0.000001 |

1 in = 2.54 cm · 1 ft = 12 in · 1 lb ≈ 0.4536 kg · 1 kg ≈ 2.2 lb · 1 L = 1,000 mL · 1 mL = 1 $cm^3$

**Conversions**
- 2.5 L = 2,500 mL · 250 mg = 0.25 g · 0.5 mg = 500 μg · 0.75 km = 750 m
- 150 lb × 0.45359237 ≈ 68.0 kg
- 5 ft 6 in = (5 × 12 + 6) in × 2.54 ≈ 167.6 cm
- °C = (°F − 32)/1.8: 98.6 °F = 37 °C; °F = 1.8 × °C + 32: 40 °C = 104 °F
- K = °C + 273.15: 37 °C = 310.15 K

Moving to a smaller unit gives a bigger number (multiply); moving to a larger unit gives a smaller number (divide).

# Common Traps

**TRAP**
- Pulmonary arteries carry deoxygenated blood and pulmonary veins oxygenated blood: arteries are defined by direction (away from the heart), not by $O_2$.
- Tendons join muscle to bone; ligaments join bone to bone.
- Insulin lowers blood glucose; glucagon raises it. Calcitonin lowers blood $Ca^{2+}$; PTH raises it.
- Mitosis gives two identical diploid cells; meiosis gives four different haploid cells.
- RNA has uracil, not thymine; codons are read on mRNA.
- Aa × Aa gives a 3 : 1 phenotype ratio but a 1 : 2 : 1 genotype ratio.
- Antibiotics act on bacteria, not on viruses.
- Lower pH = more acidic; one pH unit = tenfold change in $[H^+]$.
- Balance with coefficients, never by changing subscripts.
- The independent variable is changed; the dependent variable is measured.
- Precise is not the same as accurate.

Written with AI assistance; content checked and approved by the seller. · A study aid. It complements your courses and reference books; it doesn't replace them. · TEAS is a registered trademark of ATI (Assessment Technologies Institute). Not affiliated with or endorsed by ATI.